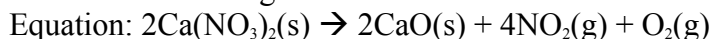


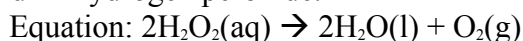
Gas Volumes

1. 10.0 g of calcium nitrate is heated at until it fully decomposes. The gas collected is cooled to rtp.

- a) the volume of nitrogen dioxide evolved
- b) the volume of oxygen evolved
- c) the total volume of gas evolved



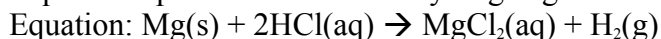
2. Calculate the volume of oxygen produced by the decomposition of 30 cm³ of 0.1 moldm⁻³ hydrogen peroxide.



3. Lead (IV) oxide dissolves in concentrated hydrochloric acid according to the following equation: $\text{PbO}_2(\text{s}) + 4\text{HCl}(\text{aq}) \rightarrow \text{PbCl}_2(\text{s}) + \text{Cl}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$
Starting with 37.2 g of lead (IV) oxide, calculate:

- a) the volume of 12 moldm⁻³ HCl needed to completely dissolve it
- b) the mass of PbCl₂ produced
- c) the volume of chlorine produced.

4. What mass of magnesium, and what volume of 2.0 moldm⁻³ hydrochloric acid, will be required to produce 100 cm³ of hydrogen gas?



5. 0.52 g of sodium was added to 100 cm³ of water. Calculate:

- a) The volume of hydrogen evolved.
- b) The concentration of the sodium hydroxide solution produced, assuming the volume of water does not change.

